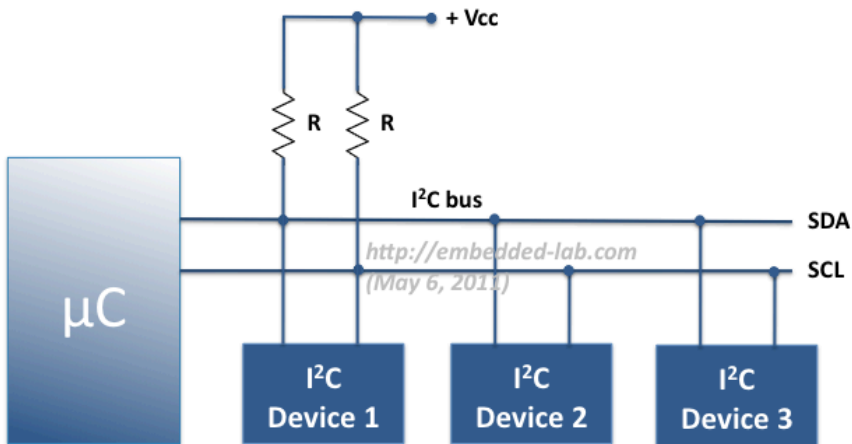


I2C Communication Steps

In multiple master-multiple slave config. the master which pulls the SDA line first will get to communicate with the slave. [Only write this line for multiple master-multiple slaves]

1. $\overline{SDA} = 0$ and sends start condition
2. $\overline{SCL} = 0$ and establishes synchronization
3. Sends 7-10 bits of address through the $\overline{SDA}$ line to find the slave and the operation as well. Read = 1 / write = 0.
4. Once the Slave is found, slave sends acknowledgement ($\overline{ACK}=0$)
5. The data frames (8 bits) are sent through the $\overline{SDA}$ line along with the acknowledgement
6. Once the data is received $\overline{ACK}=0$ is sent from the slave
7. $\overline{SCL} = 1$ and breaks synchronization
8. $\overline{SDA} = 1$ and sends stop condition



Multiple devices on common I2C bus

UART Communication Steps

1. The microprocessor sends 8 bits of data to the data bus buffer of device D1 through parallel lines.
2. From the data bus buffer the data is then sent to the Tx-Rx module of the device.
3. Device D1 then creates a data packet with start bit 0, the actual data of 8 bits, parity bit and stop bit.

*Here, if even number of 1s' in the data then parity bit = 0
if odd number of 1s' in the data then parity bit = 1*

4. Gradually sends the data to Device D2 bit by bit using serial line.
5. Device D2 accumulates the data packet once again and checks for errors.
6. If the parity bit matches, then sends the data (8 bits only) to the processor of Device D2.

